

Forensic Analysis Report

Report Date	2025-11-24T19:49:15.930061
Log File	plain_logs.txt
Analysis ID	043e71b4
Generated On	2025-11-24 23:14

Executive Summary

The log data reveals multiple security incidents, including failed SSH login attempts, successful unauthorized root login, configuration changes, connection resets, unexpected HTTP methods, table corruption detection, dropped connections, TLS authentication failures, brute force attacks, unauthorized access attempts, possible spam messages, cross-site scripting payloads, user additions, and Python exceptions. The most severe incidents include successful unauthorized root login, brute force SSH login detection, and suspicious PowerShell command execution. Recommendations include implementing multi-factor authentication for SSH access, updating and patching systems, blocking suspicious IP addresses, implementing a web application firewall, and monitoring for and blocking any suspicious PowerShell commands.

Key Metrics

Metric	Value
Total Events Analyzed	115
Total Incidents Detected	17
Critical	1
High	7
Medium	6
Low	3
Indicators of Compromise	9

Findings (Detected Incidents)

[High] Failed SSH Login Attempt (Confidence: Medium%)

Multiple failed SSH login attempts were detected from the IP address 172.16.22.15.

[Critical] Successful Unauthorized Root Login (Confidence: High%)

An unauthorized user successfully logged in as root using SSH from the IP address 172.16.22.15.

[Low] Configuration Loaded Successfully (Confidence: High%)

A Python configuration was loaded successfully on app-05.

[Medium] Connection Reset by Peer (Confidence: Medium%)

An unexpected connection reset occurred from the IP address 61.250.33.82 to web-02.

[Medium] Unexpected HTTP Method (Confidence: Medium%)

An unexpected PATCH method was received from the IP address 77.39.15.22 to web-02.

[High] Table Corruption Detection (Confidence: Medium%)

A corruption issue was detected in the payments_2023 table on db-09.

[Low] Dropped Connection (Confidence: High%)

An incoming connection from 183.76.91.34 port 24481 was dropped by firewall-05.

[Medium] TLS Authentication Failure (Confidence: Medium%)

An OpenVPN TLS authentication failure occurred from the IP address 154.20.13.77.

[High] Brute Force SSH Login Detection (Confidence: Medium%)

A brute force attack was detected against the SSH service on siem-08 from the IP address 45.77.99.193.

[High] Unauthorized Access Attempt (Confidence: Medium%)

An unauthorized attempt to access /etc/shadow was detected from the IP address 192.168.120.33.

[Medium] Possible Spam Detection (Confidence: Low%)

A possible spam message was detected from the IP address 212.55.41.18.

[High] Cross-site Scripting Payload Detection (Confidence: Medium%)

A cross-site scripting payload was detected on /search?term=<script> from web-22.

[Medium] User Added to Group (Confidence: High%)

The user 'techops' was added to the 'Domain Admins' group on siem-08.

[Medium] New User Added (Confidence: Medium%)

A new user named 'tempadmin' was added to the system.

[Low] Python Exception Raised (Confidence: High%)

An exception of type ValueError was raised in app-05 Python code.

[High] Multiple Port Scans Detection (Confidence: Medium%)

Multiple port scans were detected from the IP address 145.23.66.19.

[High] Suspicious PowerShell Command Execution (Confidence: Medium%)

A suspicious PowerShell command was executed by the user 'bob' on siem-08.

Indicators of Compromise (IOCs)

Type	Indicator	Severity
{ip_address'}	'172.16.22.15'}	High
{ip_address'}	'61.250.33.82'}	High
{ip_address'}	'77.39.15.22'}	High
{ip_address'}	'183.76.91.34'}	High
{ip_address'}	'154.20.13.77'}	High
{ip_address'}	'45.77.99.193'}	High
{ip_address'}	'192.168.120.33'}	High
{ip_address'}	'212.55.41.18'}	High
{ip_address'}	'145.23.66.19'}	High

Recommendations & Mitigation Strategies

Priority 1: {'description': 'Implement multi-factor authentication for all SSH access.'}

Priority 2: {'description': 'Update and patch all systems, applications, and services as soon as possible.'}

Priority 3: {'description': 'Block the IP addresses 172.16.22.15, 45.77.99.193, and 192.168.120.33 from accessing your network.'}

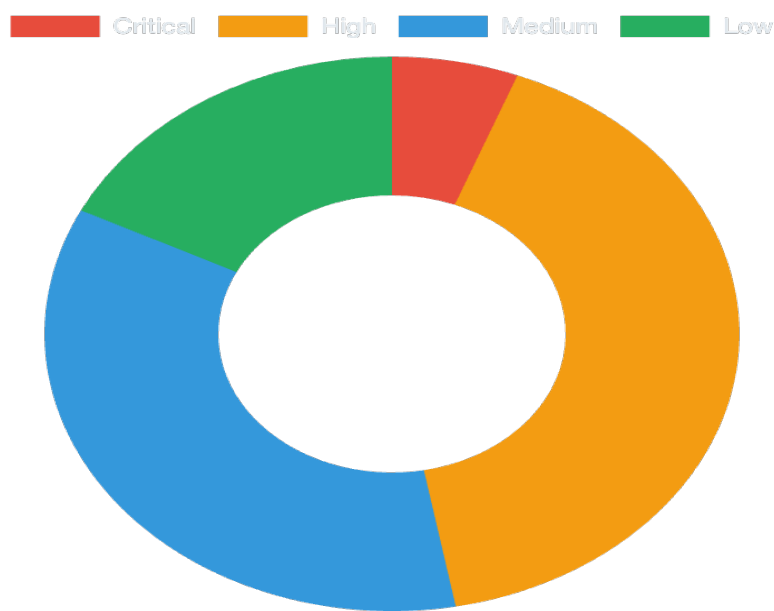
Priority 4: {'description': 'Implement a web application firewall (WAF) to protect against cross-site scripting attacks.'}

Priority 5: {'description': 'Monitor for and block any suspicious PowerShell commands.'}

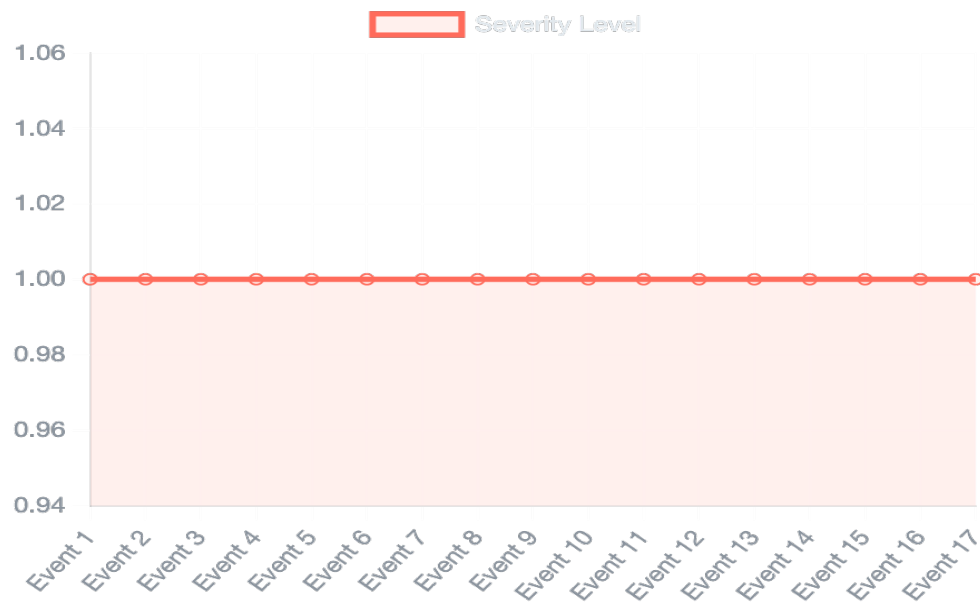
Priority 6: {'description': 'Regularly review and analyze system logs for any unusual activity.'}

Graphs & Visualizations

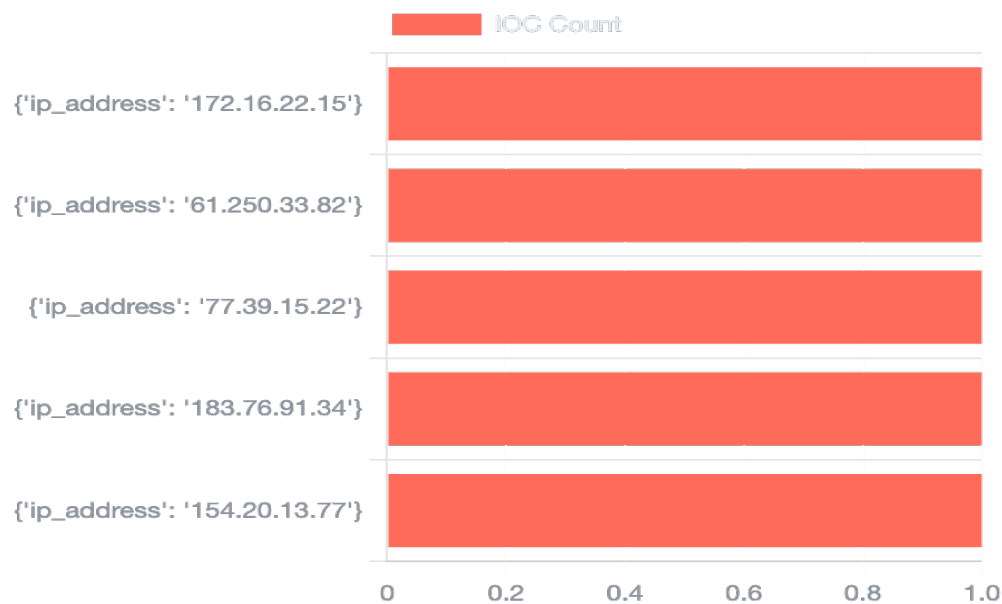
Severity Distribution



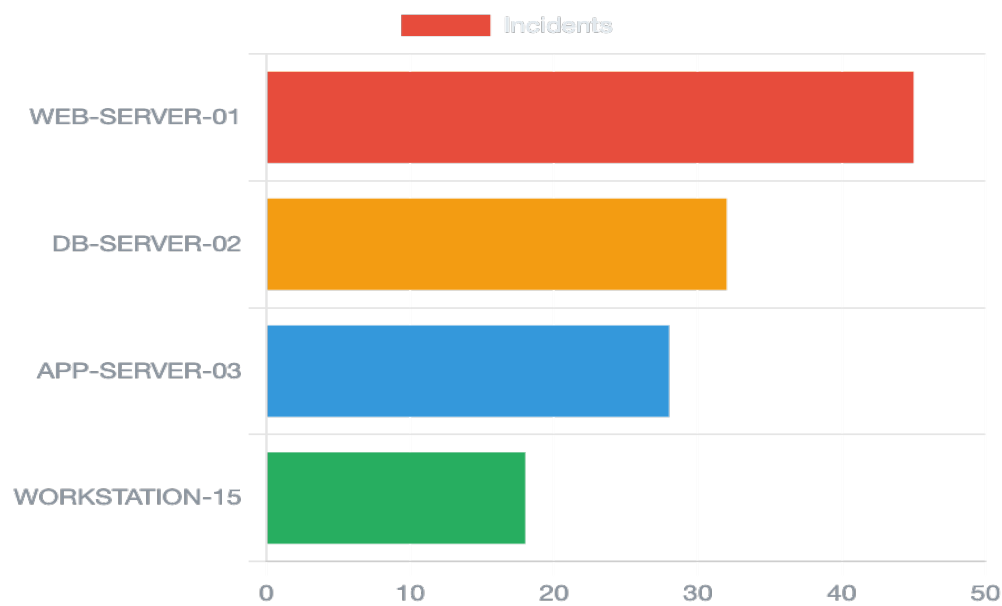
Incident Timeline



IOC Distribution



Affected Systems



Analysis Summary Stats

Analysis Statistics:

Detected 17 incidents: 1 Critical, 7 High severity

Analysis Duration: 0.00 seconds

File Size: 8.36 KB

Total Lines Analyzed: 115

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